

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
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MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
NETWORK SECURITY (CYS360)
(6TH SEMESTER, 3RD Year) LAB EXPERIMENT # 3

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Lab # 3 To Configure Firewall NAT Server and Source NAT

Objective: The objective of this lab is to configure Source NAT and NAT Server

Duration: 3 Hours

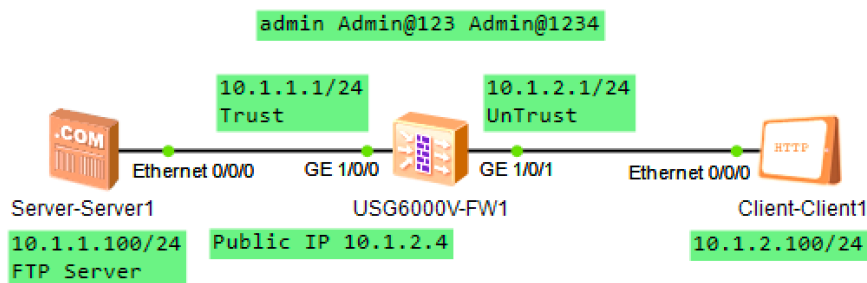
Expected Outcomes: By the end of this lab session, students will be able to:

1. Translation of addresses between networks (NAT)
2. Implement security policy

Rubrics

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT
SCORE			

*Note: Student performance will be assessed based on this rubric during lab sessions. **Topology:***



Scenario

As shown in Figure, the client (source) is shown to the right of the firewall, and the server (destination) is shown to the left of the firewall. To enable a client on the private network to communicate with a server on the another private network, we need to configure NAT Server on the firewall and map private addresses to a public address.

Tasks

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Step 1 Preparing the environment (Firewall)

The device is running!

Login authentication

Username: admin Password:

The password needs to be changed. Change now? [Y/N]: y

Please enter old password: Admin@123

Please enter new password: Admin@1234

Please confirm new password: Admin@1234

Info: Your password has been changed. Save the change to survive a reboot.

```
*****
*           Copyright (C) 2014-2018 Huawei Technologies Co., Ltd.           *
*           All rights reserved.                                           *
*           Without the owner's prior written consent,                    *
*           no decompiling or reverse-engineering shall be allowed.       *
*****
```

Configuration:

<USG6000V1>

<USG6000V1>system-view

Enter system view, return user view with Ctrl+Z.

[USG6000V1]

[USG6000V1]interface GigabitEthernet 1/0/0

[USG6000V1-GigabitEthernet1/0/0]ip address 10.1.1.1 24

[USG6000V1-GigabitEthernet1/0/0]q

[USG6000V1]interface GigabitEthernet 1/0/1

[USG6000V1-GigabitEthernet1/0/1]ip address 10.1.2.1 24

[USG6000V1-GigabitEthernet1/0/0]q

[USG6000V1]

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Step 2 Configure IP address to Client and Server

Server1

Basic Config | Server Info | Log Info

MAC Address: 54-89-98-CE-6A-7C (Format:00-01-02-03-04-05)

IPv4 Config

Local Address: 10 . 1 . 1 . 100 Subnet Mask: 255 . 255 . 255 . 0

Gateway: 0 . 0 . 0 . 0 DNS: 0 . 0 . 0 . 0

Ping Test

IPv4 Address: 0 . 0 . 0 . 0 Times: Send

Local State: Device boot ping success: 0 failed: 0

Save

Client1

Basic Config | Client Info | Log Info

MAC Address: 54-89-98-83-65-6D (Format:00-01-02-03-04-05)

IPv4 Config

Local Address: 10 . 1 . 2 . 100 Subnet Mask: 255 . 255 . 255 . 0

Gateway: 0 . 0 . 0 . 0 DNS: 0 . 0 . 0 . 0

Ping Test

IPv4 Address: 0 . 0 . 0 . 0 Times: Send

Local State: Device boot ping success: 0 failed: 0

Save

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Step 3 Configure Firewall Zone (Optional)

```
[USG6000V1] display zone
2020-04-22 18:57:25.660
local priority is 100
interface of the zone is (0):
# trust priority is 85
interface of the zone is (1):
    GigabitEthernet0/0/0
# untrust
priority is 5 interface of the
zone is (0):
#
dmz
priority is 50 interface of
the zone is (0):
#
[USG6000V1]
[USG6000V1] firewall zone name huawei
[USG6000V1-zone-huawei]
[USG6000V1-zone-huawei] set priority 45
[USG6000V1-zone-huawei] q
[USG6000V1] display zone
2020-04-22 19:00:52.260
local priority is 100
interface of the zone is (0):
# trust priority is 85
interface of the zone is (1):
    GigabitEthernet0/0/0
# untrust
priority is 5 interface of the
zone is (0):
#
dmz
priority is 50 interface of
the zone is (0):
#
huawei
priority is 45
```

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interface of the zone is (0):

#

[USG6000V1]

Step 4 Add Firewall Interfaces to Security Zones

[USG6000V1] firewall zone trust

[USG6000V1-zone-trust] add interface GigabitEthernet 1/0/0

[USG6000V1-zone-trust] display this

2020-04-22 19:00:54.280

firewall zone

trust set priority

85

add interface GigabitEthernet0/0/0 add

interface GigabitEthernet1/0/0

#

return

[USG6000V1-zone-trust]q

[USG6000V1]

[USG6000V1] firewall zone untrust

[USG6000V1-zone-untrust] add interface GigabitEthernet 1/0/1

[USG6000V1-zone-untrust] display this

2020-04-22 19:00:58.350

#

firewall zone untrust

set priority 5

add interface GigabitEthernet1/0/1

#

return

[USG6000V1-zone-untrust]q

[USG6000V1]

Step 5 Configure Security Policy

[USG6000V1] security-policy

[USG6000V1-policy-security] rule name 1

[USG6000V1-policy-security-rule-1] source-zone untrust

[USG6000V1-policy-security-rule-1] destination-zone trust

[USG6000V1-policy-security-rule-1] action permit

[USG6000V1-policy-security-rule-1] service ftp

[USG6000V1-policy-security-rule-1] display this

2020-04-22 21:47:27.700

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```
# rule name 1
source-zone untrust
destination-zone trust
service ftp action
permit
# return
[USG6000V1-policy-security-rule-1]q
[USG6000V1-policy-security]q
[USG6000V1]
Step 6 Configure Source NAT
[USG6000V1] nat address-group 1
[USG6000V1-address-group-1] section 10.1.1.50 10.1.1.60
[USG6000V1-address-group-1] display this
2020-04-22 21:58:24.330
#
nat address-group 1 0 mode
pat
section 0 10.1.1.50 10.1.1.60
#
return
[USG6000V1-address-group-1]q
[USG6000V1] nat-policy
[USG6000V1-policy-nat] rule name A
[USG6000V1-policy-nat-rule-A] source-address 10.1.2.0 mask 255.255.255.0
[USG6000V1-policy-nat-rule-A] action source-nat address-group 1
[USG6000V1-policy-nat-rule-A] display this
2020-04-22 22:02:13.420
# rule name
A
source-address 10.1.2.0 mask 255.255.255.0 action
source-nat address-group 1
# return
[USG6000V1-policy-nat-rule-A] quit
[USG6000V1-policy-nat] quit
[USG6000V1]
<USG6000V1> save
```

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Step 7 Configure NAT Server

The **nat server** command defines a mapping table of internal servers so that external users can access internal servers through address and port translation. **nat server name protocol { tcp | udp } global global-address global-port inside hostaddress [host-port]**

```
[USG6000V1] nat server ftp-server protocol tcp global 10.1.2.4 ftp inside 10.1.1.100 ftp
```

```
[USG6000V1] display nat server all-systems
```

2020-04-23 00:40:03.720

Server in private network information:

```
Total 1 NAT server(s) server
name : ftp-server
id    : 0          zone    : ---
global-start-addr : 10.1.2.4      global-end-addr  : 10.1.2.4
inside-start-addr : 10.1.1.100    inside-end-addr  : 10.1.1.100
global-start-port : 21(ftp)       global-end-port  : 21      inside-
start-port : 21(ftp)             inside-end-port  : 21      globalvpn
: public      insidevpn : public      vsys      : public
protocol : tcp          no-revers : 0          interface :
---          unr-route  : 0          description : ---
nat-disable : 0
```

```
[USG6000V1]
```

```
[USG6000V1] display firewall server-map
```

2020-04-24 05:58:30.560

Current Total Server-map : 2

Type: Nat Server, ANY -> 10.1.2.4:21[10.1.1.100:21], Zone:---, protocol:tcp Vpn:
public -> public

Type: Nat Server Reverse, 10.1.1.100[10.1.2.4] -> ANY, Zone:---, protocol:tcp
Vpn: public -> public, counter: 1

Try following commands to troubleshoot errors

```
[USG6000V1] display firewall session table
```

```
[USG6000V1] display interface brief
```

```
[USG6000V1] display memory-usage
```

```
[USG6000V1] display alarm active
```

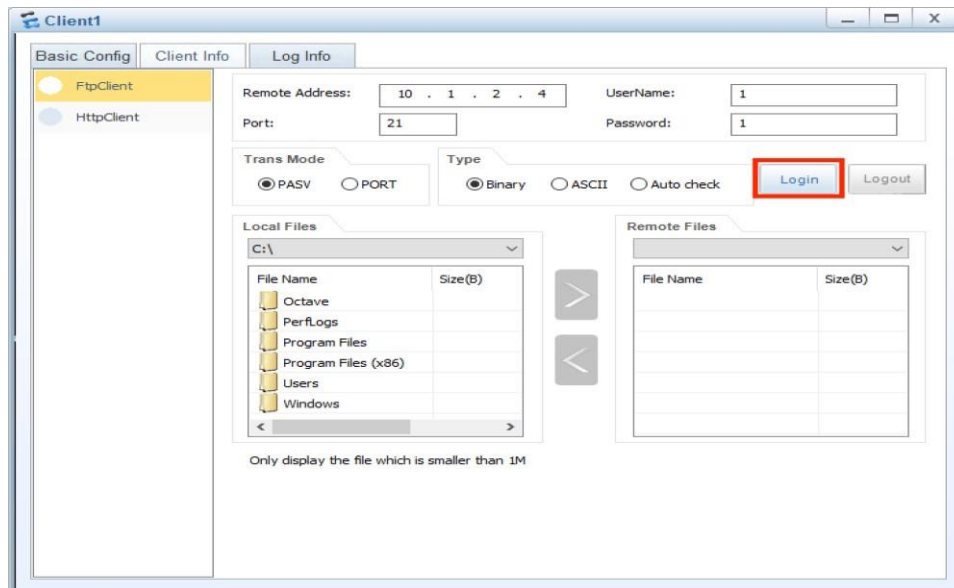
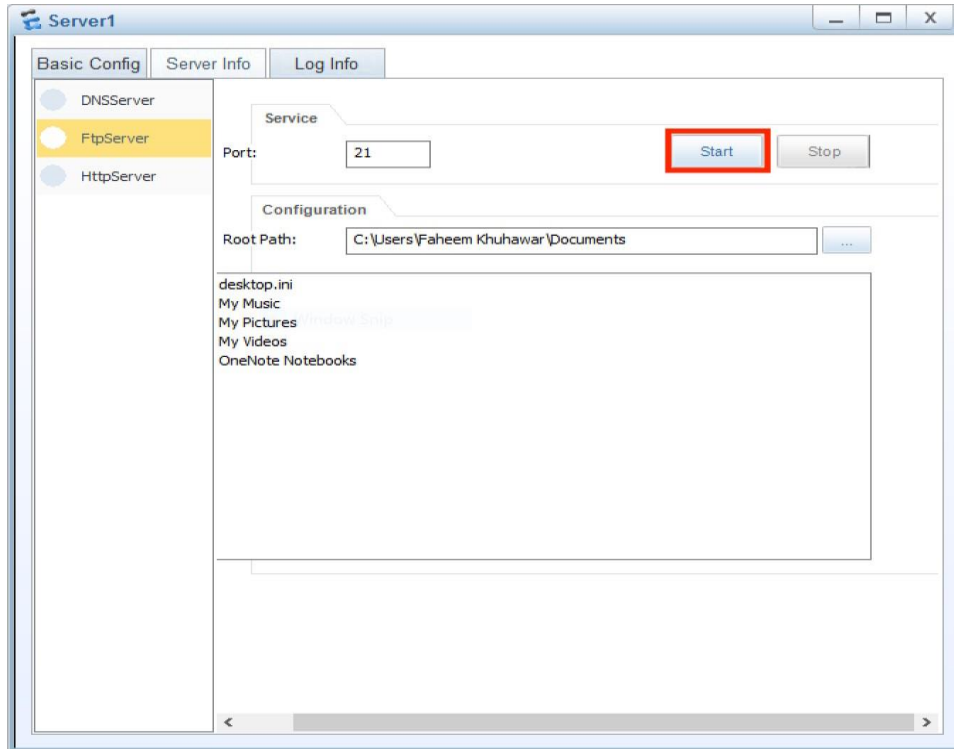
```
[USG6000V1] display alarm history
```

```
[USG6000V1] display logbuffer
```

```
[USG6000V1] display trapbuffer
```

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Step 8 Configure FTP Server and Client



Questions

1. What is the purpose of Source NAT?

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To **change the source IP of outgoing traffic** so internal devices can access external networks.

2. What are the basic steps to configure Source NAT on a Huawei USG6000V firewall?

- Create a **NAT address group** (pool of public IPs).
- Create a **NAT policy**.
- Define the **source network** and **action** (use address group).
- Apply and verify the NAT rule.

3. How does Source NAT help conserve public IPv4 addresses?

Many internal devices can share **a few public IPs** when accessing the internet.

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4. What is the difference between Source NAT and Destination NAT?
- **Source NAT:** Changes **source IP** for outgoing traffic.
 - **Destination NAT:** Changes **destination IP** for incoming traffic (port forwarding).
5. What is the difference between **Easy-IP**, **NAPT**, and **No-NAPT** modes in Huawei Source NAT configuration?
- **Easy-IP:** Automatic NAT for simple setups.
 - **NAPT:** Many internal IPs share **one public IP** using port translation.
 - **No-NAPT:** Each internal IP uses a **one-to-one public IP** (no port translation).
6. In Huawei USG6000V, what is the purpose of a “**NAT policy**” and how is it applied?
- NAT policy **defines which traffic gets NATed** and which NAT action to take.
 - Applied by **matching source/destination and assigning a NAT action**.
7. Which command is used in the **CLI** of Huawei USG6000V to create a Source NAT rule?

nat-policy

rule name <rule_name>

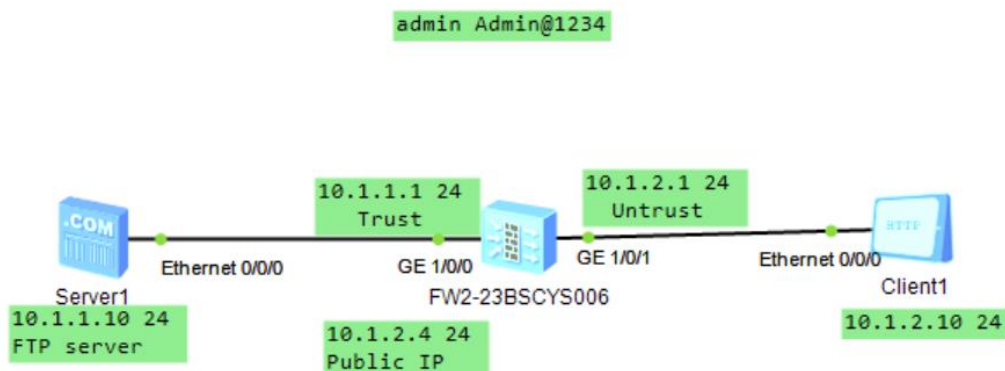
source-address <internal_network> mask <subnet_mask>

action source-nat address-group <group_number>

Attach all the screenshots of the configuration. The Firewall name should be your roll number like FW1-22BSCYS001

Note: Upload the **.topo and **.cfg** files on MS Teams Assignment.**

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```
interface GigabitEthernet0/0/0
undo shutdown
ip binding vpn-instance default
ip address 192.168.0.1 255.255.255.0
alias GEO/METH
#
interface GigabitEthernet1/0/0
undo shutdown
ip address 10.1.1.1 255.255.255.0
#
interface GigabitEthernet1/0/1
undo shutdown
ip address 10.1.2.1 255.255.255.0
#
interface GigabitEthernet1/0/2
undo shutdown
#
interface GigabitEthernet1/0/3
undo shutdown
#
interface GigabitEthernet1/0/4
undo shutdown
```

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```
#
interface Virtual-if0
#
interface NULL0
#
firewall zone local
    set priority 100
#
firewall zone trust
    set priority 85
    add interface GigabitEthernet0/0/0
    add interface GigabitEthernet1/0/0
#
firewall zone untrust
    set priority 5
    add interface GigabitEthernet1/0/1
#
```

```
#
firewall detect ftp
#
    nat server ftp-server protocol tcp global 10.1.2.4 ftp inside 10.1.1.10 ftp
#
user-interface con 0
    authentication-mode aaa
user-interface vty 0 4
```

```
#
nat address-group 1 0
    mode pat
    section 0 10.1.1.50 10.1.1.60
#
multi-linkif
    mode proportion-of-weight
#
right-manager server-group
```

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```
#
security-policy
rule name untrust-trust
  source-zone untrust
  destination-zone trust
  service ftp
  action permit
#
auth-policy
#
traffic-policy
#
policy-based-route
#
nat-policy
rule name A
  source-address 10.1.2.0 mask 255.255.255.0
  action source-nat address-group 1
#
```

Client1

Basic Config Client Info Log Info

☒ FtpClient
☐ HttpClient

Remote Address: 10 . 1 . 2 . 4 UserName: 1
Port: 21 Password: 1

Trans Mode Type
☒ PASV ☐ PORT ☒ Binary ☐ ASCII ☐ Auto check Login Logout

Local Files Remote Files

C:\Usman\UNI\NS (Practical)\

File Name	Size(B)
..	
FW-GUI	
LAB-01	
LAB-02	
LAB-03	
NS - Lab 03 SNAT.pdf	528060

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File Name	Size(B)
FW-GUI	
LAB-01	
LAB-02	
LAB-03	
NS - Lab 03 SNAT.pdf	528060
NS - Lab 1.pdf	341185

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